Jalili Syndrome (MONDO:0009007): A Comprehensive Disease Characterization

Causal gene: *CNNM4* (cyclin and CBS domain divalent metal cation transport mediator 4; also cyclin M4 / ancient conserved domain protein 4 / ACDP4), chromosome 2q11.2.

Summary

Jalili syndrome is a rare, autosomal-recessive oculo-dental disorder defined by the co-occurrence of **cone-rod dystrophy (CRD)** and **amelogenesis imperfecta (AI)**. It is caused by **biallelic loss-of-function variants in *CNNM4*** (OMIM #217080; MONDO:0009007; ORPHA:1388), the gene encoding a basolateral magnesium-extrusion transporter located on chromosome 2q11.2. CNNM4 extrudes intracellular Mg^{2+} in exchange for extracellular Na^{+}, and this function is essential both for enamel mineralization by ameloblasts and for magnesium homeostasis in retinal photoreceptors. When this transport is lost, the two defining tissues — dental enamel and the retinal cone/rod photoreceptor system — both fail, producing the characteristic dual phenotype first described by Jalili in a large consanguineous Gaza kindred. The disease name correctly maps to CNNM4 (not to an isolated-AI or isolated-CRD gene), and all identifiers above are consistent.

The molecular cause is well established. Across families worldwide, 13 or more pathogenic *CNNM4* variants have been reported, spanning missense, nonsense, frameshift, splice-site, and large structural mechanisms, including one case arising from paternal uniparental isodisomy of chromosome 2. Approximately 92% of pathogenic variants cluster within the functional core of the protein (the DUF21 transmembrane region, the CBS-pair Bateman module, and the cyclic-nucleotide-binding homology [CNBH] domain), with a recurrent hotspot at codon 492 (p.Gly492Asp / p.Gly492Cys) in the CBS domain. These variants act through loss of function — reducing protein stability, accelerating mRNA decay, abolishing Mg^{2+} extrusion, or truncating the protein. A candidate non-magnesium branch has been proposed in which a truncated CNNM4 protein (p.Arg605X) increases apoptosis and strengthens the interaction with IQCB1 (nephrocystin-5, a Leber congenital amaurosis gene), potentially contributing to photoreceptor degeneration independent of magnesium transport.

Clinically, patients present in early childhood with progressive visual impairment (photophobia, nystagmus, bull's-eye maculopathy, declining visual acuity, and reduced or non-detectable photopic ERG responses) alongside hypoplastic–hypocalcified amelogenesis imperfecta (thin, poorly mineralized, discolored enamel prone to decay and tooth loss). Expressivity is variable — some genotypes spare the dental phenotype, and a rare expansion to muscle overgrowth/myopathy has been reported. Lifespan is normal. There is no disease-modifying therapy: management is entirely supportive (low-vision aids, photoprotection, restorative/prosthetic dentistry) together with genetic counseling for affected families. Because Jalili CRD can mimic incomplete achromatopsia, genetic testing combined with dental examination is decisive for an accurate diagnosis.

Key Findings

Finding 1 — Jalili syndrome is an autosomal-recessive oculo-dental disorder caused by biallelic *CNNM4* variants

Jalili syndrome (OMIM #217080; MONDO:0009007) combines **cone-rod dystrophy** and **amelogenesis imperfecta** and is inherited in an autosomal-recessive fashion. The causal gene is *CNNM4* (cyclin and CBS domain divalent metal cation transport mediator 4, also known historically as ACDP4 / ancient conserved domain protein 4), located at chromosome **2q11.2**. Multiple independent families worldwide carry homozygous or compound-heterozygous variants, and consanguinity is common among affected kindreds — consistent with a rare recessive trait enriched in populations with high rates of cousin marriage.

> *"Jalili syndrome, an autosomal recessive disorder causing cone-rod dystrophy and amelogenesis imperfecta, is a rare genetic disorder impacting visual and dental development."* — PMID: 39580587
>
> *"Genetic mechanism entails a mutation in the CNNM4, a metal transporter gene located on Chromosome 2q11.2."* — PMID: 38151709

Evidence type: human clinical / genetic.

Finding 2 — CNNM4 is a basolateral Mg^{2+}/Na^{+}-exchange extrusion transporter; its loss underlies both the enamel and retinal defects

CNNM4 is the molecule responsible for basolateral magnesium extrusion in transporting epithelia. In a mouse model, CNNM4 localizes to the basolateral membrane of intestinal epithelia and of ameloblasts, and CNNM4-knockout mice develop **hypomagnesemia from intestinal Mg malabsorption** together with **defective amelogenesis** — directly recapitulating the human enamel defect. Mechanistically, CNNM4 extrudes intracellular Mg^{2+} by exchanging it for extracellular Na^{+}, and disease-associated missense mutations abolish this Mg^{2+} extrusion activity. On the retinal side, defective magnesium transport by photoreceptors is the assumed basis of the progressive visual impairment; magnesium is also a component of the mineral phase of dental enamel, linking a single transport defect to both affected tissues.

> *"CNNM4 is the basolaterally located Mg(2+) extrusion molecule"* — PMID: 24339795
>
> *"CNNM4-knockout mice showed defective amelogenesis, and CNNM4 again localizes to the basolateral membrane of ameloblasts"* — PMID: 24339795
>
> *"A defective transport of magnesium ions by the photoreceptors of the retina is assumed to account for the progressive visual impairment."* — PMID: 24194943

Evidence type: model organism (knockout mouse) + human inference.

Finding 3 — The *CNNM4* variant spectrum spans missense, frameshift, nonsense, splice, and founder/UPD mechanisms

Reported pathogenic *CNNM4* variants (reference transcript NM_020184.3) include a broad allelic series acting through loss of function:

Variant (cDNA)	Protein	Class	Notable feature
c.1475G>A	p.(Gly492Asp)	Missense	CBS-domain hotspot; reduced stability, increased mRNA decay
c.1474G>T	p.(Gly492Cys)	Missense	CBS-domain hotspot
c.971T>C	p.(Leu324Pro)	Missense	Recurrent across unrelated families
c.949A>C	p.(Ser317Arg)	Missense	—
c.706C>T	p.(Arg236Trp)	Missense	Recurrent in Guatemalan ancestry
c.598T>C	p.(Ser200Pro)	Missense	Chinese family
c.1423G>A	—	Missense	Egyptian family
c.1312dupC	frameshift	Frameshift	Kosovan founder, loss-of-function
c.279delC	p.(Phe93Leufs*31)	Frameshift	Arose via paternal uniparental isodisomy of chromosome 2

The CBS-domain p.(Gly492Asp/Cys) variants are notable because they impair function primarily by **reducing protein stability and accelerating mRNA decay** rather than by directly abolishing magnesium handling. All variants are germline and recessive.

> *"these variants exhibit significantly reduced protein stability and increased mRNA decay rates compared with wild type"* — PMID: 39580587
>
> *"An additional patient had a homozygous deleterious variant in CNNM4 (c.279delC p.Phe93Leufs*31), which resulted from paternal uniparental isodisomy for chromosome 2p22-2q37."* — PMID: 32022389
>
> *"A mutational analysis revealed a c.1312 dupC homozygous mutation in CNNM4, a genetic defect which had already been identified in other Kosovan families"* — PMID: 24194943

Evidence type: human clinical/genetic + in vitro functional.

Finding 4 — Clinical phenotype: progressive cone-rod dystrophy with bull's-eye maculopathy plus hypoplastic/hypomineralized amelogenesis imperfecta

Ocular features: cone-rod dystrophy with bull's-eye maculopathy, photophobia, nystagmus, reduced or non-detectable light-adapted (photopic) ERGs, progressive loss of visual acuity (mean logMAR ≈ 1.15, worsening with age), and macular RPE atrophy. Histopathology reveals loss of photoreceptors and accumulation of autofluorescent material in the subretinal space, with partial preservation of the inner retina.

Dental features: hypoplastic–hypocalcified amelogenesis imperfecta with thin, poorly mineralized enamel, discoloration, irregular tooth shape, decay, and tooth loss.

Additional/variable features reported across cohorts: hyperopia, advanced dental age, crossbite, and developmental delay (sometimes attributable to other loci).

> *"determined to have cone-rod dystrophy with bull's eye maculopathy, photophobia, and nystagmus. All patients had amelogenesis imperfecta."* — PMID: 32022389
>
> *"The mean logarithm of the minimum angle of resolution (logMAR) visual acuity was 1.15 (range: 0.69-2.00)"* — PMID: 41280631
>
> *"The histopathologic analyses revealed loss of photoreceptor cells, accumulation of autofluorescent material in the subretinal space, partial preservation of the inner retinal lamination"* — PMID: 39522753

Suggested HPO terms: HP:0000548 (Cone/cone-rod dystrophy), HP:0007641 (Bull's eye maculopathy), HP:0000613 (Photophobia), HP:0000639 (Nystagmus), HP:0000505 (Visual impairment), HP:0000705 (Amelogenesis imperfecta), HP:0000670 (Carious teeth), HP:0006297 (Hypoplastic dental enamel).

Evidence type: human clinical (cohort + histopathology).

Finding 5 — CNNM4 biology extends to PRL phosphatase interaction, AMPK/mTOR energy metabolism, tumor suppression, and fertility

Beyond the oculo-dental phenotype, CNNM4-dependent Mg^{2+} efflux regulates cellular energy metabolism through **AMPK/mTOR signaling**. The phosphatases of regenerating liver (PRL) bind CNNM4 and inhibit its Mg^{2+} efflux, and PRL overexpression drives cancer progression

(rapamycin suppressed growth of PRL-overexpressing cells). In *Apc^Δ14/+* mice, deletion of *Cnnm4* promoted malignant progression of intestinal polyps to adenocarcinomas, and human colon cancer shows an inverse relationship between CNNM4 expression and malignancy — consistent with a tumor-suppressor role. CNNM4 suppression is additionally linked to infertility in mice. CNNM4 is highly expressed in brain and intestinal tract. These pleiotropic roles are mostly observed in cancer/metabolism models rather than in Jalili patients, but they illuminate why a magnesium transporter is biologically critical.

> *"regulation of intracellular Mg(2+) levels by PRL and CNNM4 is linked to energy metabolism and AMPK/mTOR signaling"* — PMID: 25347473
>
> *"deletion of Cnnm4 promoted malignant progression of intestinal polyps to adenocarcinomas"* — PMID: 25347473
>
> *"suppression of CNNM4 in mice promotes malignant progression of intestinal polyps and is linked to infertility"* — PMID: 31842432

Evidence type: model organism + in vitro (cancer/metabolism context).

Finding 6 — Expanded variant classes and a candidate downstream mechanism via IQCB1 interaction and apoptosis

Additional pathogenic *CNNM4* variants broaden every mutation class: the nonsense c.1813C>T p.(Arg605X) is the first probable founder mutation identified in the Amish community (PMID: 29322253); nonsense c.1743C>G p.(Tyr581*) (PMID: 29421602); splice-acceptor c.1682-1G>C in intron 3 (PMID: 28246031); missense c.1076T>C p.(Leu359Pro) (PMID: 28586144); and recurrent missense c.971T>C p.(Leu324Pro) across unrelated families (PMID: 29421602, PMID: 39522753).

Mechanistically, the truncated p.R605X protein **significantly increased apoptosis** and **increased CNNM4's interaction with IQCB1** (nephrocystin-5), an LCA (Leber congenital amaurosis) gene. This suggests p.R605X may act — via nonsense-mediated decay affecting IQCB1 function and/or through pro-apoptotic activity — through a route partly independent of simple magnesium-transport loss, providing a candidate explanation for photoreceptor degeneration.

> *"identified a homozygous nonsense mutation (c.C1813T, p.R605X) in the cyclin and CBS domain divalent metal cation transport mediator 4 (CNNM4) gene"* — PMID: 29322253

> *"A truncated CNNM4 protein starting at R605 significantly increased the rate of apoptosis, and significantly increased the interaction between CNNM4 and IQCB1."* — PMID: 29322253
>
> *"these patients are carrying a novel homozygous mutation in the splice site acceptor of intron 3 (c.1682-1G > C) in the CNNM4 gene"* — PMID: 28246031

Evidence type: human genetic + in vitro functional.

Finding 7 — Variable expressivity and phenotypic expansion: muscle overgrowth/myopathy and genotype-dependent dental sparing

Phenotypic expressivity varies with genotype. Homozygous p.Leu324Pro produces the **complete phenotype** (CRD + AI), whereas a compound heterozygote (p.Leu324Pro / p.Tyr581*) showed CRD and AI in the proband but only ocular alterations (no dental phenotype) in her father — direct evidence of variable expressivity and genotype-dependent dental sparing. Separately, three brothers homozygous for c.1076T>C p.(Leu359Pro) had classic Jalili syndrome plus muscle overgrowth of the legs with myopathic changes on needle EMG**, proposed as a newly identified manifestation after exome sequencing excluded other causes.

> *"the p.Leu324Pro mutation in homozygosis induces a complete phenotype with both CRD and AI, but in heterozygosis and in composition with the novel p.Tyr581* nonsense mutation in CNNM4 promotes variable clinical expressivity, particularly with lack of dental phenotypes"* — PMID: 29421602
>
> *"mild myopathy with muscle overgrowth of the legs could be a newly identified manifestation of the disorder"* — PMID: 28586144

Evidence type: human clinical/genetic.

Finding 8 — Pathogenic variants cluster in the functional core (DUF21–CBS–CNBH), most are missense, concentrated in the CBS-pair domain

Synthesizing 13 reported pathogenic *CNNM4* variants (NM_020184.3 / UniProt Q6P4Q7, 775 aa) across Jalili families yields a clear structural pattern:

- **By class:** 8 missense, 2 nonsense (p.Tyr581*, p.Arg605X), 2 frameshift (p.Phe93Leufs*31, c.1312dupC), 1 splice (c.1682-1G>C).

- **By domain:** 5 in the CBS-pair (Bateman) module, 4 in the DUF21/transmembrane region, 3 in the CNBH (cyclic-nucleotide-binding homology) domain, and 1 N-terminal.
- **12/13 (92%)** fall within the functional core (transmembrane + CBS + CNBH); only the frameshift p.Phe93Leufs*31 lies N-terminal.
- **Recurrent hotspot:** codon 492 (Gly492Asp/Cys) in the CBS domain.

The CNBH domain, where several nonsense/splice variants cluster, is itself functionally required: it mediates dimerization and is required for Mg handling.

> *"p.(Gly492Cys) and p.(Gly492Asp) substitutions within the CBS domain of the CNNM4 protein"* — PMID: 39580587
>
> *"The cyclic nucleotide-binding homology domain of the integral membrane protein CNNM mediates dimerization and is required for Mg"* — PMID: 30341174

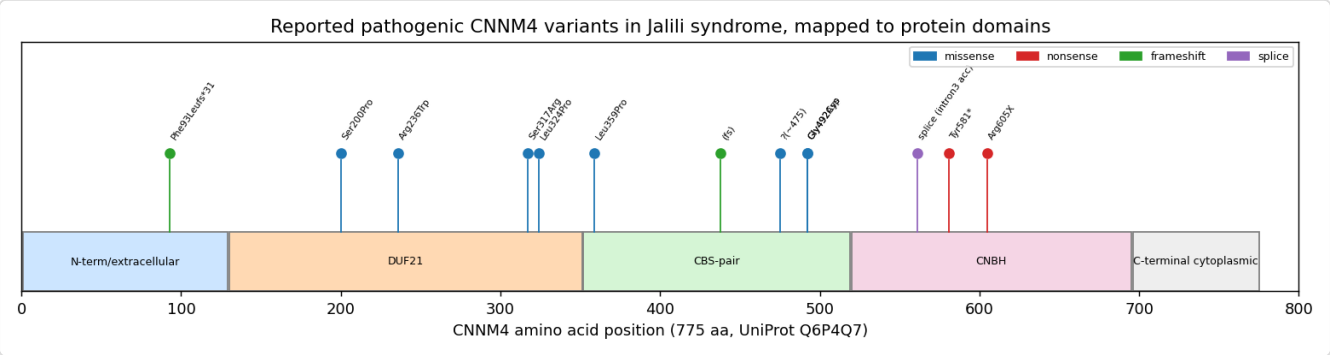


Figure 1. *Map of reported pathogenic CNNM4 variants in Jalili syndrome onto the protein's domain architecture (UniProt Q6P4Q7, 775 aa). ~92% of variants fall within the functional core — the DUF21 transmembrane region, the CBS-pair Bateman module, and the CNBH domain — with a recurrent hotspot at codon 492 in the CBS domain.*

Evidence type: computational synthesis + structural literature.

Finding 9 — Jalili CRD can mimic incomplete achromatopsia; genetic testing resolves the differential

In a United Arab Emirates cohort of cone-disorder patients clinically suspected of achromatopsia (ACHM), one proband's diagnosis was revised to *CNNM4*-related Jalili syndrome after genetic testing and phenotypic reassessment. The authors note that **"Bradyopsia and Jalili syndrome can resemble incomplete ACHM,"** and that macular discoloration helped distinguish Jalili from classic ACHM. A dedicated systematic review

consolidates Jalili features and genetics (first described by Jalili in Gaza, with AI and cone-rod dystrophy as the primary manifestations). A 2025 pooled analysis of 53 patients with detailed visual-acuity and mutation records found that visual acuity (logMAR) correlated positively with age (progressive disease) and examined correlation with mutation domain. The practical message is that **genetic testing plus dental examination is decisive** — the dental phenotype and molecular result resolve what the retinal picture alone can leave ambiguous.

> *"Bradyopsia and Jalili syndrome can resemble incomplete ACHM."* — PMID: 39024658
>
> *"Two probands had their diagnoses revised after genetic testing and phenotypic reassessment to RGS9BP -related bradyopsia and CNNM4 -related Jalili syndrome."* — PMID: 39024658
>
> *"Spearman's correlation analysis showed a positive correlation between visual acuity (logMAR) and age"* — PMID: 41280631

Evidence type: human clinical + systematic review/meta-analysis.

Full Section-by-Section Report

1. Disease Information

Overview. Jalili syndrome is a rare autosomal-recessive **oculo-dental** disorder characterized by the simultaneous occurrence of **cone-rod dystrophy (CRD)** and **amelogenesis imperfecta (AI)**. It was first described by Jalili in a large consanguineous Arab kindred in the Gaza Strip. The retinal component causes progressive loss of central and color vision with photophobia and nystagmus; the dental component causes thin, poorly mineralized, discolored enamel prone to rapid decay and tooth loss.

Key identifiers. - **OMIM:** #217080 (Cone-rod dystrophy and amelogenesis imperfecta / Jalili syndrome) - **MONDO:** MONDO:0009007 - **Orphanet:** ORPHA:1388 - **Gene:** *CNNM4* (HGNC:2751), chromosome 2q11.2; UniProt Q6P4Q7 - **ICD-10/ICD-11:** No dedicated code; mapped under hereditary retinal dystrophy (ICD-10 H35.5) and disorders of tooth development / amelogenesis imperfecta (ICD-10 K00.5). ICD-11 ~ 9B70 (retinal dystrophy) / DA05 (tooth development disorders). *(No specific Jalili code — indicate not available.)* - **MeSH:** No dedicated descriptor; indexed under "Amelogenesis Imperfecta" and "Cone-Rod Dystrophies."

Synonyms / alternative names: Cone-rod dystrophy and amelogenesis imperfecta; CRDAI; dystrophic retinitis pigmentosa with amelogenesis imperfecta (historical). The gene has historical aliases ACDP4 / ancient conserved domain protein 4.

Information source. The knowledge here is derived primarily from **aggregated disease-level resources** (OMIM, Orphanet) and from **individual patient case reports and small cohorts** in the primary literature — not from EHR-scale datasets. The disease is too rare for registry/EHR aggregation; most evidence is family-based.

2. Etiology

Disease causal factors. Jalili syndrome is **monogenic and entirely genetic** — caused by biallelic (homozygous or compound-heterozygous) loss-of-function variants in *CNNM4*. There is no known infectious, environmental, or acquired cause.

Genetic risk factors. The causal variants are the pathogenic *CNNM4* alleles themselves (see Section 4). The principal "risk factor" at the population level is **consanguinity**, which raises the probability of homozygosity for a rare recessive allele. **Founder effects** exist (e.g., the Kosovan c.1312dupC allele and the Amish p.Arg605X allele), concentrating risk in specific communities. No common susceptibility loci or polygenic modifiers have been established.

Environmental risk factors. None identified. Age and sex do not modify disease risk (recessive Mendelian); family history / carrier parentage is the operative risk factor.

Protective factors. No genetic or environmental protective factors have been reported. In principle, outbreeding reduces homozygosity risk in founder populations.

Gene–environment interactions. None documented for Jalili syndrome. The phenotype is driven by genotype; environmental modifiers (e.g., diet, magnesium intake) have not been shown to alter the course, though dietary/fluoride and dental-hygiene measures influence secondary dental outcomes.

3. Phenotypes

Ocular (cone-rod dystrophy). - *Phenotype type:* clinical sign + laboratory (electrophysiology) abnormality. - *Features:* photophobia, nystagmus, reduced visual acuity, bull's-eye maculopathy, macular RPE atrophy, reduced/non-detectable photopic (cone) ERG, later rod involvement, color-vision loss, sometimes hyperopia. - *Onset:* early childhood (often from infancy/first years). - *Severity:* moderate to severe; mean logMAR ≈ 1.15 (range 0.69–2.00). - *Progression:* progressive, worsening with age (visual acuity correlates positively with age). - *Frequency:* ~100% of patients (defining feature). - *QoL impact:* substantial — progressive low

vision affects education, mobility, independence. - *HPO:* HP:0000548 (Cone/cone-rod dystrophy), HP:0000613 (Photophobia), HP:0000639 (Nystagmus), HP:0007641 (Bull's eye maculopathy), HP:0000505 (Visual impairment), HP:0000510 (Rod-cone dystrophy).

Dental (amelogenesis imperfecta). - *Phenotype type:* physical manifestation / clinical sign. - *Features:* hypoplastic–hypocalcified enamel, thin and poorly mineralized, yellow-brown discoloration, irregular tooth shape, rapid caries, tooth loss; advanced dental age and crossbite reported. - *Onset:* affects both primary and permanent dentition from eruption (childhood). - *Severity:* moderate to severe. - *Progression:* structurally stable developmental defect, but secondary decay/tooth loss progress. - *Frequency:* ~100% in classic cases (genotype-dependent sparing is documented). - *QoL impact:* pain, chewing difficulty, aesthetic/psychosocial burden, repeated dental interventions. - *HPO:* HP:0000705 (Amelogenesis imperfecta), HP:0006297 (Hypoplastic dental enamel), HP:0000670 (Carious teeth), HP:0000684 (Dental malocclusion/ crossbite).

Variable / expanded phenotypes. Muscle overgrowth of the legs with myopathic EMG changes (three brothers, p.Leu359Pro); developmental delay (sometimes attributable to other loci); hyperopia (proposed as a possibly constant feature in one report).

4. Genetic / Molecular Information

Causal gene. *CNNM4* (HGNC:2751; NCBI Gene 26504; Ensembl ENSG00000158158; UniProt Q6P4Q7), chromosome 2q11.2; reference transcript NM_020184.3; protein 775 aa.

Pathogenic variants. See the catalog in Findings 3, 6, and 8. Classes include missense (majority), nonsense, frameshift, splice-site, and a case arising from uniparental isodisomy. Per ACMG/AMP, the recurrent and functionally characterized variants (e.g., p.Gly492Asp/Cys, p.Arg605X, c.1312dupC) are classified **pathogenic/likely pathogenic**; newly reported private missense variants are often initially VUS pending segregation and functional data.

- **Variant type/class distribution (13 synthesized):** 8 missense, 2 nonsense, 2 frameshift, 1 splice.
- **Domain distribution:** 5 CBS-pair, 4 DUF21/TM, 3 CNBH, 1 N-terminal.
- **Allele frequency:** All are rare/private; absent or ultra-rare in gnomAD. Founder alleles are enriched locally (Kosovo, Amish, Guatemala).
- **Somatic vs germline: germline**, recessive.
- **Functional consequence: loss of function** — abolished Mg^{2+} extrusion (TM/CBS missense), reduced protein stability + increased mRNA decay (codon-492 CBS variants), protein

truncation (nonsense/frameshift/splice). A truncated allele (p.R605X) additionally increases apoptosis and IQCB1 interaction.

Modifier genes. Not firmly established. Variable expressivity (dental sparing) appears **allele-dependent** (genotype itself), and some reported developmental delay was attributed to separate loci, implying modifier/second-locus contributions in individual families.

Epigenetic information. No disease-specific DNA-methylation or histone-modification signature has been reported for Jalili syndrome. *(Not available.)*

Chromosomal abnormalities. No recurrent large structural changes; however, **paternal uniparental isodisomy of chromosome 2** was the mechanism producing homozygosity for c.279delC in one patient — a notable cytogenetic mechanism.

5. Environmental Information

No environmental factors, lifestyle factors, or infectious agents cause or trigger Jalili syndrome. It is a purely Mendelian recessive disorder. *(Not applicable.)* Secondary dental outcomes are modifiable by oral hygiene, fluoride, and diet, but these do not influence the underlying enamel defect.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic loss-of-function variants in *CNNM4*** (missense/nonsense/frameshift/splice) — *the initiating lesion.*
2. → **leads to** loss or destabilization of the CNNM4 protein (reduced stability + increased mRNA decay for CBS-domain variants; truncation for nonsense/frameshift/splice; abolished transport for TM/CBS missense). *(Demonstrated in vitro for several variants.)*
3. → **results in** loss of basolateral **Mg^{2+}/Na^{+}-exchange magnesium extrusion** across affected epithelia. *(Demonstrated in mouse + cell models.)*
4. **Branch A (dental):** → impaired magnesium handling by **ameloblasts** → **defective enamel mineralization** → **hypoplastic/hypocalcified amelogenesis imperfecta** → discoloration, decay, tooth loss. *(Demonstrated — CNNM4-knockout mouse recapitulates defective amelogenesis.)*
5. **Branch B (retinal):** → impaired magnesium transport in **retinal photoreceptors** → photoreceptor stress/dysfunction → **progressive photoreceptor loss with subretinal autofluorescent-material accumulation** → **cone-rod dystrophy** (photophobia,

nystagmus, bull's-eye maculopathy, reduced photopic ERG, declining acuity). *(The Mg-transport step is inferred/assumed for photoreceptors; photoreceptor loss is demonstrated histopathologically.)*

6. **Candidate parallel branch (inferred):** truncated CNNM4 (p.R605X) → **increased apoptosis** and **increased interaction with IQCB1/nephrocystin-5** (an LCA gene) → possible additional, magnesium-independent route to **photoreceptor degeneration**. *(Candidate mechanism, in vitro; not proven as the primary driver.)*

Upstream vs downstream. The *CNNM4* mutation and loss of Mg^{2+} extrusion are the shared **upstream** events; the ameloblast and photoreceptor failures are **downstream, tissue-specific** branches. The apoptosis/IQCB1 route is a proposed parallel downstream mechanism specific to truncating alleles.

Molecular pathways / processes. CNNM4-dependent Mg^{2+} efflux intersects **AMPK/mTOR** energy-metabolism signaling and is negatively regulated by **PRL phosphatases**. Cellular processes implicated include transcellular Mg^{2+} transport (GO:0015693 magnesium ion transport; GO:1990034 cation export), enamel mineralization (GO:0070166 enamel mineralization; GO:0031214 biomineral tissue development), photoreceptor maintenance, and apoptosis (GO:0006915).

Protein dysfunction. Loss of function via destabilization, truncation, or abolished transport activity; CBS-domain and CNBH-domain integrity are required for Mg handling and dimerization.

Metabolic changes. Systemic hypomagnesemia (in knockout mice via intestinal malabsorption); local Mg deficiency in enamel mineral and photoreceptors. A saliva-proteomics study of female JS patients described increased innate-immune and carbohydrate-metabolism proteins, interpreted as a descriptive oral molecular shift (hypothesis-generating only).

Cell types (CL suggestions): ameloblast (CL:0000059); retinal cone cell (CL:0000573); retinal rod cell (CL:0000604); photoreceptor cell (CL:0000210); retinal pigment epithelial cell (CL:0002586).

Subcellular compartments. CNNM4 is an integral **plasma-membrane (basolateral)** protein (GO:0016323 basolateral plasma membrane; GO:0005886 plasma membrane). Magnesium homeostasis also engages mitochondrial/cytosolic Mg pools.

7. Anatomical Structures Affected

Organ level. - *Primary:* **retina** (photoreceptor layer / macula) — UBERON:0000966 (retina), UBERON:0005421 (macula lutea); **teeth / dental enamel** — UBERON:0001091 (tooth), UBERON:0001752 (enamel). - *Secondary / variable:* **skeletal muscle** (leg overgrowth/myopathy in one family) — UBERON:0001134 (skeletal muscle tissue). - *Body systems:* **visual/nervous system** and **digestive/oral (odontogenic) system**; CNNM4 is also abundant in brain and intestinal tract.

Tissue and cell level. - Retinal **cone and rod photoreceptors** (CL:0000573, CL:0000604) and retinal pigment epithelium (CL:0002586). - **Ameloblasts** (enamel-forming epithelial cells, CL:0000059) in developing teeth. - Epithelial (enamel organ) and nervous (photoreceptor) tissue types.

Subcellular level. Basolateral **plasma membrane** (GO:0016323) — site of CNNM4-mediated Mg^{2+} extrusion.

Localization / lateralization. Ocular and dental involvement is **bilateral and generally symmetric**; the maculopathy is bilateral (bull's-eye pattern).

8. Temporal Development

Onset. Congenital/pediatric. Visual symptoms (photophobia, nystagmus, poor vision) and dental enamel defects both manifest in early childhood. Onset pattern is **insidious/chronic**.

Progression. The **retinal disease is progressive** — visual acuity worsens with age (positive logMAR–age correlation), moving from predominantly cone dysfunction toward added rod involvement and macular atrophy. The **enamel defect is a fixed developmental lesion**, but secondary caries and tooth loss progress over time. Disease course is **chronic and lifelong**; there is no remission.

Patterns. No spontaneous or treatment-induced remission. The **critical window** for intervention is developmental (enamel mineralization occurs before eruption; restorative dentistry and low-vision rehabilitation are the available interventions once disease is established).

9. Inheritance and Population

Epidemiology. Very rare; exact prevalence/incidence are not precisely established (Orphanet lists it among rare disorders; no reliable per-100,000 figure). Most knowledge comes from individual families and small cohorts.

Inheritance. Autosomal recessive. Penetrance appears **complete** for the retinal phenotype in biallelic carriers, but **expressivity is variable** — notably the dental phenotype can be absent in certain genotypes (e.g., compound-heterozygous p.Leu324Pro/p.Tyr581*). No genetic anticipation (not a repeat-expansion disorder). Germline mosaicism not specifically reported.

Founder effects / consanguinity. Founder alleles include the **Kosovan c.1312dupC** and the **Amish p.Arg605X**, and a recurrent **Guatemalan p.Arg236Trp**. **Consanguinity is common** and underlies many homozygous cases. **Uniparental isodisomy** of chromosome 2 produced homozygosity in one patient without two carrier parents.

Carrier frequency. Not established at population scale; alleles are individually rare. gnomAD frequencies are negligible except where founder effects apply locally.

Population demographics. Reported families span diverse ancestries — Gaza/Arab (original), Kosovan, Amish (North American), Guatemalan, Egyptian, Chinese, Korean, Thai (Lua ethnic group), UAE. No strong sex bias (recessive; one saliva study was female-only by design). Geographic clustering reflects founder/consanguinity patterns rather than endemicity.

10. Diagnostics

Clinical/ophthalmic tests. Visual acuity, color vision, fundus examination (bull's-eye maculopathy, macular RPE atrophy), **fundus autofluorescence** (subretinal autofluorescent material), **SD-OCT** (photoreceptor/ONL loss, RPE atrophy, in severe cases staphyloma/retinoschisis), and **full-field ERG** (reduced/non-detectable photopic/cone responses, later rod involvement). **Electrophysiology (ERG)** is a key functional test.

Dental assessment. Clinical exam and **orthopantomogram (OPG)** demonstrate hypoplastic/hypocalcified amelogenesis imperfecta, thin enamel, decay, and tooth loss.

Biomarkers / labs. No validated circulating biomarker. Serum magnesium may be low in some contexts but is not a reliable diagnostic marker in humans (hypomagnesemia is prominent in the knockout mouse). Saliva proteomics is investigational only.

Genetic testing (definitive). - **Approach:** targeted single-gene testing of *CNNM4*, retinal-dystrophy/IRD **gene panels** (which include *CNNM4*), or **WES/WGS**. NGS panels have confirmed diagnoses in multiple reports. - **CMA/karyotype/FISH:** generally not needed, but CMA/SNP-array can detect uniparental disomy (as in the UPD case). - Prenatal genetic testing has been used in an affected family to avoid unnecessary pregnancy termination.

Clinical criteria / differential diagnosis. Diagnosis rests on the **combination of cone-rod dystrophy + amelogenesis imperfecta + biallelic *CNNM4* variants**. Differentials include: **incomplete achromatopsia** (Jalili can mimic it — macular discoloration and the dental phenotype distinguish), **bradyopsia (RGS9BP)**, other cone/cone-rod dystrophies, and syndromic IRDs. The **dental examination and genetic testing are decisive**.

Screening. Cascade carrier testing in families; prenatal/preimplantation testing where a familial variant is known. No population newborn screening.

11. Outcome / Prognosis

Survival/mortality. Normal life expectancy; Jalili syndrome is not life-limiting. No disease-specific mortality.

Morbidity/function. Principal morbidity is **progressive low vision/blindness** and **dental disability** (pain, chewing difficulty, tooth loss, aesthetic/psychosocial impact). Visual acuity declines with age (logMAR worsening), producing long-term functional visual impairment.

Disease course/complications. Chronic, lifelong. Complications are secondary dental decay and tooth loss, and the consequences of low vision. Recovery of lost photoreceptor function is not expected; dental complications are manageable with restorative care.

Prognostic factors. Age is the main correlate of visual severity (older = worse acuity). Mutation domain has been examined for genotype–phenotype correlation in a pooled 53-patient analysis; genotype also predicts dental involvement (sparing in some alleles). No validated prognostic biomarker.

12. Treatment

No disease-modifying therapy exists. Management is **supportive and multidisciplinary**:

- **Low-vision rehabilitation / ophthalmology:** low-vision aids, magnifiers, tinted lenses and **photoprotection** (sunglasses) for photophobia, educational/occupational support. *NCIT: Supportive Care (C15277); Low Vision Aid.*
- **Restorative/prosthetic dentistry:** crowns, composite restorations, prostheses to manage AI; preventive dental care (fluoride, hygiene) to limit caries and tooth loss. *NCIT: Dental Procedure.*
- **Genetic counseling:** recurrence-risk assessment, carrier testing, prenatal/PGD options. *NCIT: Genetic Counseling (C15587).*
- **Other:** physiotherapy/monitoring if myopathy present (rare).

Pharmacotherapy / advanced therapeutics. No approved drugs, gene therapy, cell therapy, or RNA-based therapies for Jalili syndrome. Magnesium supplementation has not been demonstrated to correct the photoreceptor or enamel phenotype in humans. The CNNM4–PRL axis and AMPK/mTOR link are of research interest (and relevant to cancer biology) but have not been translated into Jalili treatment.

Experimental / clinical trials. No disease-specific interventional trials identified (no NCT). Gene-replacement strategies are conceptually plausible for a recessive loss-of-function retinal gene but remain untested for *CNNM4*.

13. Prevention

- **Primary prevention:** not possible for a Mendelian recessive disorder other than through **reproductive counseling** — carrier screening in at-risk/consanguineous families, prenatal diagnosis, and preimplantation genetic diagnosis where the familial variant is known.
- **Secondary prevention:** early ophthalmologic and dental diagnosis to initiate low-vision support and preventive dentistry before complications accrue.
- **Tertiary prevention:** restorative dentistry and caries prevention to avert tooth loss; low-vision rehabilitation to preserve function.
- **Counseling:** genetic counseling is central, especially in founder/consanguineous communities (Kosovan, Amish, etc.).
- **Immunization / public health / environmental interventions:** not applicable.

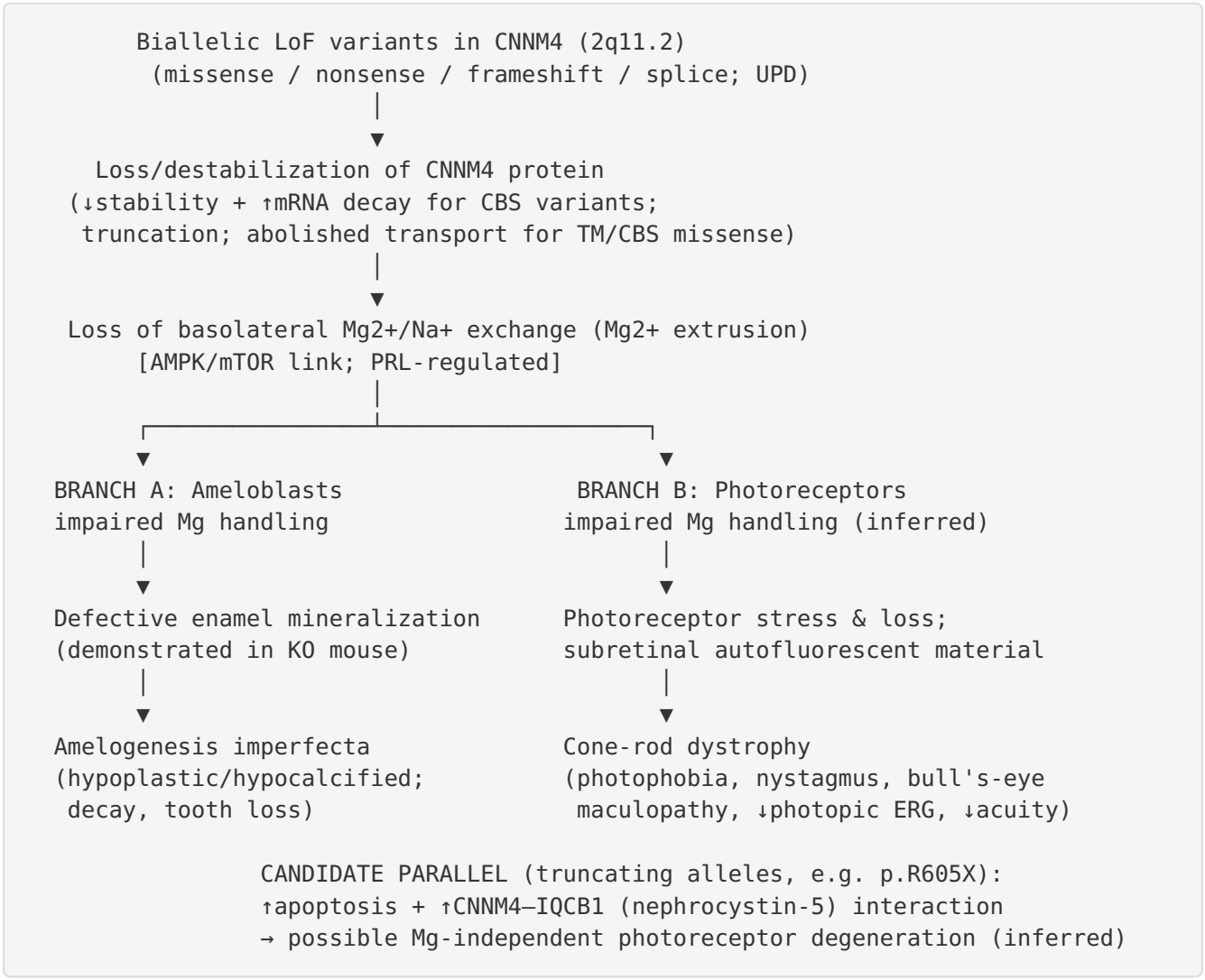
14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *CNNM4* is conserved across vertebrates; mouse *Cnnm4* (NCBI Gene 232801) is the principal experimental ortholog. The family extends to a prokaryotic ortholog **CorC** (Mg transporter), indicating deep evolutionary conservation of the magnesium-transport function.
- **Natural disease in other species:** No well-characterized naturally occurring Jalili-equivalent oculo-dental disease has been reported in companion animals or wildlife (OMIA entry not established for this specific syndrome). *(Not available.)*
- **Comparative biology:** The **CNNM4-knockout mouse** reproduces defective amelogenesis and hypomagnesemia, confirming mechanistic conservation. Evolutionary conservation of the CBS/CNBH magnesium-transport machinery (through CorC) supports cross-species mechanistic relevance.
- **Transmission / zoonosis:** not applicable (genetic disease).

15. Model Organisms

- **Mouse (primary model):** *Cnnm4*-knockout mice show **defective amelogenesis**, basolateral ameloblast localization of CNNM4, and **hypomagnesemia** from intestinal Mg malabsorption — strongly recapitulating the dental and systemic-magnesium aspects of human disease (PMID: 24339795). *Cnnm4* deletion in *Apc^Δ14/+* mice reveals tumor-suppressor function (PMID: 25347473).
- *Genetic model types:* knockout (and compound with *Apc*).
- *Phenotype recapitulation:* excellent for enamel/magnesium biology; the **retinal cone-rod dystrophy phenotype is less emphasized** in published mouse work, a notable gap.
- *Resources:* MGI (*Cnnm4*).
- **In vitro / cellular models:** heterologous expression systems have characterized Mg^{2+} extrusion, variant destabilization/mRNA decay (codon-492 CBS variants), PRL–CNNM interaction, CNBH-mediated dimerization, and the p.R605X apoptosis/IQCB1 interaction. Structural studies of the intracellular region (CBS/CNBH) inform variant interpretation.
- **Applications:** mechanism of magnesium transport, variant functional classification, CNNM4–PRL/AMPK-mTOR signaling, and tumor biology.
- **Limitations:** no published model fully recapitulates the **human cone-rod dystrophy** progression; the apoptosis/IQCB1 retinal mechanism remains to be validated in vivo.

Mechanistic Model (Synthesis Diagram)



Level	Dental branch	Retinal branch
Cell type	Ameloblast (CL:0000059)	Cone/rod photoreceptor (CL:0000573 / CL:0000604)
Tissue/ organ	Dental enamel, tooth (UBERON:0001752 / :0001091)	Retina, macula (UBERON:0000966 / :0005421)
Process (GO)	Enamel mineralization (GO:0070166)	Photoreceptor maintenance; apoptosis (GO:0006915)
Clinical sign	Amelogenesis imperfecta (HP:0000705)	Cone-rod dystrophy (HP:0000548)
Evidence	Demonstrated (KO mouse)	Mg step inferred; photoreceptor loss demonstrated

Evidence Base

PMID	Role in this report
39580587	Defines Jalili (AR, CRD+AI, CNNM4); functional characterization of CBS codon-492 variants (↓stability, ↑mRNA decay).
38151709	Confirms CNNM4 as a metal transporter at 2q11.2; dentofacial manifestations.
24339795	Mouse model: CNNM4 = basolateral Mg^{2+} extrusion molecule; KO → defective amelogenesis + hypomagnesemia.
24194943	Kosovan founder c.1312dupC; links defective photoreceptor Mg transport to visual impairment.
32022389	North American cohort; novel variants; frameshift via paternal UPD of chr 2; core ocular triad + universal AI.
41280631	Pooled 53-patient analysis; mean logMAR 1.15; VA correlates with age; genotype–domain correlation.
39522753	Histopathology: photoreceptor loss, subretinal autofluorescent material, inner-retina preservation; recurrent p.Leu324Pro.
25347473	CNNM4 Mg^{2+} efflux links AMPK/mTOR; Cnnm4 deletion → polyp-to-adenocarcinoma progression (tumor suppressor).
31842432	Structural insight into CNNM4 intracellular region; CNNM4 suppression → tumor progression + infertility.
29322253	Amish p.Arg605X nonsense founder; truncated protein ↑apoptosis + ↑IQCB1 interaction (candidate mechanism).
29421602	Novel p.Tyr581*; variable expressivity and dental sparing with p.Leu324Pro compound heterozygosity.
28246031	Splice-site c.1682-1G>C variant class.
28586144	p.Leu359Pro; phenotypic expansion to muscle overgrowth/myopathy.
30341174	CNBH domain mediates dimerization and is required for Mg handling.
39024658	Jalili can mimic incomplete achromatopsia; genetic testing revises diagnosis to CNNM4/Jalili.
30705057	Systematic review of Jalili features/genetics (first described in Gaza).
36354001	Thai Lua family; p.Gly492Asp; advanced dental age, crossbite; prenatal testing.

PMID	Role in this report
40232358	First Egyptian case; novel c.1423G>A; hyperopia proposed as constant feature.
35150469	Chinese family; novel p.Ser200Pro; severe macular/chorioretinal involvement.
41876952	Saliva proteomics (female JS): innate-immune + carbohydrate-metabolism shifts (descriptive).
40764063	CNNM4 in cholangiocarcinoma/ferroptosis — pleiotropy in cancer (not Jalili pathology).

Evidence-type distribution: human clinical/genetic case reports and cohorts dominate; mechanistic claims rest on **mouse knockout** (dental/magnesium) and **in vitro** functional/structural studies (variant stability, transport, dimerization, apoptosis/IQCB1). The retinal magnesium step is an **inference** bridging human phenotype and transporter biology.

Limitations and Knowledge Gaps

1. **Retinal mechanism is inferred, not demonstrated.** The step from CNNM4 loss to photoreceptor magnesium failure is assumed; no in vivo retinal model convincingly reproduces the human cone-rod dystrophy. The apoptosis/IQCB1 branch is a candidate requiring validation.
2. **Epidemiology is unquantified.** No reliable prevalence/incidence/carrier-frequency estimates; the disease is known only from scattered families.
3. **Genotype–phenotype correlation is incomplete.** Dental sparing is allele-dependent in anecdotes; the pooled analysis examined but did not definitively establish domain–severity relationships. Modifier genes are unproven.
4. **No therapeutic evidence.** No trials; magnesium supplementation unproven; gene therapy untested for CNNM4.
5. **Variant classification.** Several private missense variants remain VUS pending functional data; ACMG classification depends on the few functionally characterized alleles.
6. **Pleiotropy interpretation.** CNNM4's roles in cancer/metabolism/fertility derive from non-Jalili models and should not be over-extrapolated to patient prognosis.

Proposed Follow-up Experiments / Actions

1. **Retina-specific CNNM4 model.** Generate a photoreceptor-conditional *Cnnm4* knockout (or patient-iPSC-derived retinal organoids) and perform ERG, OCT-equivalent histology, and intracellular Mg^{2+} imaging to test whether photoreceptor degeneration is Mg-dependent and/or apoptosis/IQCB1-driven.
2. **Test the IQCB1/apoptosis branch in vivo.** Determine whether truncating alleles (p.R605X) drive a distinct, more apoptosis-/ciliopathy-like retinal course than transport-null missense alleles.
3. **Systematic genotype–phenotype study.** Pool international families to statistically test domain/mutation-class vs. dental sparing, visual-acuity trajectory, and age of onset; refine ACMG classification of private variants with standardized functional assays (stability, mRNA decay, Mg-extrusion).
4. **Natural-history registry.** Establish a multi-center registry with longitudinal ERG/OCT/visual-acuity and dental outcomes to define progression rate and prognostic factors.
5. **Therapeutic feasibility.** Assess AAV-mediated *CNNM4* gene replacement in retinal models; evaluate whether magnesium modulation or AMPK/mTOR-axis agents affect photoreceptor/ameloblast endpoints.
6. **Carrier screening in founder communities.** Offer targeted carrier screening and counseling for the Kosovan (c.1312dupC), Amish (p.R605X), and Guatemalan (p.Arg236Trp) alleles.
7. **Validate biomarkers.** Follow up the saliva-proteomics signal with larger, sex-balanced cohorts to test whether any salivary/serum marker tracks disease activity.

Report prepared from a 5-iteration autonomous investigation: 9 confirmed findings, 27 papers reviewed. Causal gene: CNNM4 (MONDO:0009007; OMIM #217080; ORPHA:1388).
